CSC311: Introduction to Machine Learning

Project Report

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1 Data Exploration

Dataset Overview

- **Dataset Composition**: The dataset contains 870 survey responses collected from 290 distinct students, with each respondent evaluating all three sports (Soccer, Baseball, and Basketball). In this dataset, there are 14 columns representing the survey questions and responses.
- **Class Balance**: The target labels are perfectly balanced (290 observations per sport overall; exactly 203 per sport in the 70% training split). No class rebalancing or oversampling is needed.

Data Splitting

- **Respondent Isolation (`id`)**: The respondent identifier (`id`) was used to perform grouped partitioning. All responses from a given student were kept within the same dataset split to prevent data leakage arising from respondent-specific vocabulary, preferences, or writing styles.
- **Train-Test-Validation Split**: The 290 respondent IDs are partitioned into:
  - **Training split (70%)**: 203 respondents (609 observations) used for exploratory analysis, distribution inspection, and feature engineering.
  - **Validation split (15%)**: 43 respondents (129 observations) reserved strictly for hyperparameter tuning.
  - **Internal test split (15%)**: 44 respondents (132 observations) held out for final evaluation of the best model.

  Following partitioning, the validation and test sets were excluded from all exploratory analysis. Feature engineering, imputation parameters, standardization statistics, and text vocabularies were learned exclusively from the training set.

Features Types

Table 1: Feature categories in the dataset.

Feature Category	Variables
Numerical	difficulty_rating
Likert-scale	strategy_importance, injury_risk_comfort, individual_brilliance
Multi-select categorical	play_location, play_season, associated_music
Free-text	associated_foods_drinks, associated_color, associated_song, emotional_response
Semi-structured text	acceptable_price
Target	label

Features Distributions

The distribution of the difficulty ratings follow roughly a normal distribution but appear skewed for baseball and basketball. The Likert-scale features are mostly unimodal, with a slight skew towards higher values for strategy importance and injury risk comfort. Multi-select categorical features show a wide variety of responses, with some locations (e.g. outdoor spaces), music genres (e.g. hip-hop or rap), seasons (e.g. summer) being more popular than others, namely winter and classical/jazz. Free-text features exhibit high variability in length and content, with some respondents providing detailed descriptions while others leave fields blank. Ticket price willingness exhibits extreme positive skewness (+10.5), spanning from $0 to $10,000.

Missing Values, Outliers, and Inconsistencies

Blank responses are rare overall (∼1.70%), with the song feature exhibiting the highest vacancy (∼90 entries missing). Rather than discarding sparse text rows, a binary indicator (`is_missing`) is appended to each text feature, allowing models to leverage deliberate omission as a predictive signal (e.g., respondents unable to name a song for less mainstream sports). Imputation is also used to handle missing values in numerical and Likert-scale features, with mean imputation for difficulty rating and neutral-value imputation for Likert features.

Outliers are present in the acceptable price feature, with some respondents reporting extreme willingness to pay. A $\log(1 + x)$ transformation is applied to reduce skewness and mitigate outlier influence. Inconsistencies are also present in this feature with some respondents reporting prices with dollar signs, commas, etc. These are stripped retaining only the numeric value during preprocessing.

Feature Preprocessing

Table 2: Feature preprocessing summary.

Feature Group	Preprocessing	Rationale
Respondent ID	Excluded from predictors; used for grouped partitioning	Prevents overfitting (IDs are not generalizable).
Difficulty Rating	Mean imputation ($\mu = 5.48$); z-score standardization	Removes scale disparity and handles missing values.
Likert Features	Monotonic mapping (1–5); neutral imputation (3)	Preserves ordinal structure while avoiding unnecessary dimensionality.
Multi-select Features	Lowercasing; delimiter splitting; multi-hot encoding	Represents multiple simultaneous selections without imposing ordering.
Text Features	Lowercasing; punctuation removal; TF–IDF vectorization	Downweights common words and emphasizes discriminative sport-specific terms.
Acceptable Price	Regex removal of "$" and commas; conversion of "thousand" expressions; median imputation; $\log(1 + x)$ transformation	Produces numeric values and reduces extreme positive skewness.
Target Label	Map Soccer → 0, Baseball → 1, Basketball → 2	Creates machine-readable class labels.

2 Methodology

Data splitting and validation We reserve 15% of respondent IDs as an internal test set, 15% as a validation set, and 70% for model training. On the remaining respondents, we will use five-fold grouped cross-validation for hyperparameter tuning. All three responses from each respondent will remain together throughout this process. Imputation values, scaling parameters, categorical vocabularies, and text vocabularies will be learned separately from each fold's training data. The internal test set will remain unused during feature selection and hyperparameter tuning. Model 1: Regularized softmax regression Softmax regression will predict the probabilities of the three sports from a combination of structured features and TF-IDF text features. It is appropriate as a relatively simple model for a dataset with many potentially sparse text features. L2 regularization will help control overfitting. We will tune the regularization parameter, compare unigrams with unigrams plus bigrams, and vary the minimum document frequency used to retain text features. We will inspect model coefficients to identify which responses contribute most strongly to each prediction. Model 2: Random forest A random forest will model nonlinear relationships and interactions between responses. For example, the relationship between a season response and a sport may depend on the locations selected by the respondent. We will first evaluate the forest using structured features, then test whether adding a restricted text vocabulary improves validation performance. We will tune maximum tree depth, minimum samples per leaf, and the number of features considered at each split. Candidate settings could include 100 and 300 trees, maximum depths of 5, 10, and unrestricted depth, and minimum leaf sizes of 1, 3, and 5. Model 3: Naive Bayes Multinomial Naive Bayes will provide a probabilistic text baseline using nonnegative word-count or TF-IDF features from the free-text questions. It is computationally inexpensive and can combine many weak word-level signals. We will tune the smoothing parameter over values such as 0.1, 0.5, 1, and 2, and compare word counts with TF-IDF. Its conditional-independence assumption may limit performance when words or survey responses are strongly related. Fair comparison and feature experiments All model families will use the same respondent-based folds and receive hyperparameter tuning before comparison. We will compare structured-only, text-only, and combined representations where appropriate to determine how much each feature group contributes. The supplied kNN implementation can serve as an additional reference baseline. We will also report a constant-class baseline, which achieves 33.3% accuracy on this balanced dataset. Evaluation We will report accuracy, macro-F1, per-class precision and recall, and confusion matrices. Accuracy measures overall correctness, while macro-F1 gives each sport equal importance and helps reveal uneven performance across classes. We will summarize the mean and standard deviation of cross-validation scores. Our primary selection metric will be mean validation accuracy, with macro-F1 and consistency across folds informing close comparisons.